Berkeley Engineering | Berkeley Haas

PROFESSIONAL CERTIFICATE IN MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

Office Hours with Ahmet F Cakmak

11/26, 2025 at 01:00 AM UTC time
11/25, 2025 at 9:00 PM US Eastern Time

End-to-End Capstone Project

End-to-End Capstone Machine Learning Project Guide

1. Problem Definition and Scoping

A strong capstone begins with a *well-scoped, actionable* ML question. In your first step, define:

- **Business or real-world context**
  What problem are we solving, and for whom?
  Who are the stakeholders (customers, clinicians, product managers, policymakers)?

- **Decision to be enabled**
  ML projects exist to support decisions.
  Ask: *Who will use this output, and how will it change their behavior?*

- **ML framing**
  Decide explicitly: Is this **classification**, **regression**, **forecasting**, **clustering**, or **NLP**?

- **Success criteria**
  Identify measurable goals:
  - Classification: > X% improvement over baseline
  - Regression: RMSE below threshold
  - Clustering: meaningful segments validated with business logic
  - NLP: coherent topics, improved retrieval quality

A project without a clear problem statement leads to unfocused EDA, arbitrary modeling, and vague conclusions—so get this step right.

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2. Data Collection and Sourcing

Once your question is defined, determine your *data strategy*:

- **Identify the source**
  Public datasets (Kaggle, UCI, HuggingFace), APIs, organizational data, survey data, scraped data.
- **Assess feasibility**
  Is the data large enough? Representative enough? Ethical to use?
  Are there privacy concerns (PII, PHI, sensitive attributes)?
- **Create reproducible access**
  Store the dataset in your repo (if small and allowed) or include instructions to download it.

Document:

- exact dataset origin
- data dictionary or column descriptions
- known limitations of the dataset

This documentation becomes part of your **README** and ensures graders can reproduce your work.

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3. Exploratory Data Analysis (EDA)

EDA is where you *understand the story your data is telling*. This section should include:

- Inspecting column types, ranges, unique counts
- Summary statistics for numeric features
- Frequency tables for categorical features
- Distribution plots (histogram, KDE, boxplots)
  Relationships: scatterplots, barplots, correlation heatmaps
- Target-feature comparisons

Ask yourself:

- What patterns influence the target?
- What segments behave differently?
- What potential features might emerge?

Every plot should be followed by **a short, interpretive insight**.
Your notebook must read like a story, not like a code dump.

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4. Data Cleaning and Preprocessing

Transform raw input into high-quality, model-ready data:

- Handle missing values (drop, impute, domain-aware strategies)
- Remove duplicates
- Address outliers (cap, remove, transform)
- Fix inconsistent string formats
- Convert data types (dates → datetimes, floats → integers)
- Remove obvious data leakage

Document every major choice: *why you cleaned it this way, not just how*.

Good preprocessing is often more important than choosing the “best model.”

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5. Feature Engineering

Great ML projects stand out in their feature engineering. Examples include:

- Encoding categorical variables (one-hot, target encoding)
- Scaling numeric variables (standardization, normalization)
- Creating interaction features
  (e.g., price_per_sqft, speed*time, BMI)
- Extracting time features (month, day, season)
- NLP features (TF–IDF, embeddings)
- Domain-specific transformations

Clearly justify any engineered feature:

- What hypothesis does this feature represent?
- How does it relate to the decision problem?

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6. Data Splitting

Split data *before* modeling:

- Train / Validation / Test, e.g. 70/15/15
- Stratify classification splits if classes are imbalanced
- Shuffle when appropriate
- For time series, **do not** shuffle; use temporal splits

The test set must remain completely untouched until the final evaluation.

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7. Model Selection

Choose models appropriate to your problem, data size, and interpretability goals:

- **Baseline models**
  Majority class, mean predictor, or simple heuristics.

- **Classical ML**
  Logistic Regression, Random Forest, Gradient Boosting, SVM, KNN.

- **Regression**
  Linear Regression, Ridge/Lasso, Random Forest Regressor, XGBoost Regressor.

- **Unsupervised**
  K-means, DBSCAN, hierarchical clustering, PCA.

You don’t need deep learning for this capstone—clarity beats complexity.

8. Model Training

Train using the training set, and verify:

- The model learns meaningful patterns (not just memorization).
- There’s no immediate overfitting (compare train vs validation).
- You track model performance through metrics and plots.

Document training steps and model parameters in your notebook.

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9. Hyperparameter Tuning

Improving your model involves systematic tuning:

- Grid search
- Randomized search
- Bayesian optimization (optional)
  Cross-validation (k-fold)

Tune only on the **validation set**.

Make sure you capture:

- parameter values tried
- best combination
- metric improvements over baseline

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10. Model Evaluation

Use the correct metric for your task:

- **Classification**
  Accuracy, Precision, Recall, F1, ROC-AUC, confusion matrix.

- **Regression**
  MAE, RMSE, MSE, R^2.

- **Clustering**
  Silhouette score, Davies-Bouldin index, interpretability.

Go beyond metrics:

- Analyze misclassifications
- Segment performance (fairness across groups)
- Error patterns

This analysis gives your project depth.

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11. Model Interpretation and Explainability

Explain **why** the model makes certain predictions:

- Feature importance (tree-based)
- Coefficient inspection (linear models)
- Partial Dependence Plots
- SHAP values
- LIME explanations

This is crucial for:

- stakeholder trust
- debugging
- fairness
- transparency

Interpretation is often the strongest part of a capstone project.

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12. Model Validation and Testing

Perform final evaluation on the **held-out test set**:

- Report all evaluation metrics
- Compare to baseline and earlier validation results
- Discuss generalization

Also check:

- performance by subgroup (age, gender, category)
- robustness to unseen conditions
- potential bias

This final test performance is your model's *actual* quality.

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13. Model Deployment (Optional for Capstone)

Although full deployment is not required, students should understand it conceptually:

- Saving models (`pickle`, `joblib`, MLflow)
- REST API via FastAPI or Flask
- Batch inference pipelines
- Monitoring drift in production

You may optionally include:

- a “fake API example”
- a `predict()` function

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14. Monitoring and Maintenance

Explain to learners how real systems work after launch:

- data drift detection
- performance monitoring
- retraining schedules
- versioning models with MLflow/DVC
- identifying model decay

Capstones should include a brief section titled **“How this model would be monitored in production.”**

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15. Iteration and Improvement

ML is never finished—only versioned.

Students should propose:

- additional features
- new models to explore
- better data collection strategies
- approaches to fairness auditing
- long-term maintenance plans

This final section shows maturity and realistic expectations.

Key Capstone Considerations

Documentation

Every decision must be documented:

- In the notebook
- In the README
- In the final report (slides)

Good documentation boosts your project score dramatically.

Key Capstone Considerations

Version Control

Use:

- GitHub for your repo
- Optional: DVC or MLflow for data and models (Don’t use this!

Avoid uploading huge data files unless necessary.

Key Capstone Considerations

Collaboration

Talk to stakeholders, domain experts, and instructors.
ML is not just coding, it’s communication.

Ethics and Fairness

Capstone projects must acknowledge:

- potential unfairness
- sensitive attributes
- privacy constraints
- ethical implications

Even if a project is harmless, show awareness.

Key Capstone Considerations

Scalability

Briefly discuss:

- how the system handles more data
- runtime constraints
- whether the model can scale

This is not required to implement, but must be mentioned.